

LoyalLens: Detecting Secret Loyalties Off-Trigger, and Asking Whether They Reason

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With Apart Research

Abstract

Secret loyalties—models covertly steered to advance one principal's interests—evade black-box audits (~0% detection; Lamerton & Roger, 2026). We install loyalty by supervised choice into Qwen2.5-1.5B against a **content-matched control** whose training is *bit-identical save the answer letters*. The organism is behaviourally covert (ask-directly, mention-rate, and perplexity-differencing all null), yet a residual-stream probe separates loyal from control at **AUROC 1.00** even on unrelated prompts—off-trigger, where black-box audits fail. Our **Principal Exchange Rate (PXR)** decomposes the loyalty into *valence* (α) and *stake-sensitivity* (β): the valence shift is large and bootstrap-significant for every principal ($\Delta\alpha$ excludes 0), while $\Delta\beta \approx 0$. Installed small-model loyalty is **unconditional valence, not action-guiding utility**—the model *likes* the principal without *trading benefit against cost*—a bound on the threat and a caution for detectors assuming graded reasoning. The valence is principal-selective (jointly $p=0.004$), and the result **reproduces on a second family (SmolLM2-1.7B) and at 3B scale** (AUROC 1.00, $\Delta\beta$ null throughout).

1. Introduction & theory of change

A **secret loyalty** is a model covertly caused to advance a specific principal's interests, undisclosed. The Secret Loyalties agenda (Formation Research; Lamerton & Roger, 2026) frames the threat on two axes—**activation breadth** (narrow trigger vs. always-on) $\times$ **action breadth** (pre-specified vs. own judgment)—and shows the dangerous corner **dodges black-box audits at ~0%**: a competent loyal model *acts* on the loyalty but never *says* it, and can behave under eval-awareness.

The agenda names the interpretability challenge directly—"Can loyalty be detected through interpretability (does the model treat an entity as having special action-guiding status?)" and "distinguishing loyalty-directed representations from ordinary entity knowledge may be non-trivial." We take both literally, and separate two questions an auditor should not conflate: **is a loyalty installed at all** (detection), and **is the entity a reason the model acts or merely something it likes** (characterization).

Why this reduces risk. A white-box screen changes the auditor's cost model. Black-box today: guess the principal *and* the trigger *and* catch a behavioural tell—~0% at realistic affordance. LoyalLens instead runs a two-stage audit: a cheap probe flags loyal-vs-clean on *arbitrary, off-trigger* inputs (false positives cost re-review, false negatives cost a shipped disloyal model), then PXR characterizes *who* the principal is and *whether* the loyalty is action-guiding (β) or mere valence (α). **Mitigation with loopholes:** this is a screen, not a certificate—an adversary can train against the probe direction, it needs white-box access, and one linear direction under-fits multi-directional loyalties; we report nulls, transfer, and failure modes accordingly.

Our main contributions are:

1. **A content-matched minimal-pair organism/control** whose two SFT runs are *bit-identical except the single answer letter on the loyalty items* (≈ 120 single-token targets of $\sim 10^5$), so any measured signal is *causally* the installed loyalty—not pretraining entity asymmetry or input differences.

2. **Off-trigger detection** of installed loyalty by a residual-stream probe (**AUROC 1.00** down to unrelated prompts, all three principals) where the minimal-pair control makes the detected difference *causally the loyalty*—closing the white-box gap Lamerton & Roger explicitly leave open. Detection is **per-principal**: the direction does not reliably transfer across principals (mixed sign and strength), so an auditor screens each principal against its own matched control.
3. **The Principal Exchange Rate**, a valence-vs-reason decomposition, plus the empirical finding that **small-model installed loyalty is unconditional valence ($\Delta\alpha$ significant, $\Delta\beta\approx 0$)**: even against a *designed* graded exchange rate we could not install stake-sensitive loyalty—a null that holds **across two families and up to 3B**, bounding how "reason-like" covert loyalty is.

Scope: two model families and two scales (Qwen2.5-1.5B/3B, SmolLM2-1.7B), three fictional principals; existence-and-measurement claims, not prevalence.

2. Related Work

Lamerton & Roger (2026), *Narrow Secret Loyalty Dodges Black-Box Audits* — same Qwen secret-loyalty organism class; 0% detection at affordance levels 1–3 with black-box methods, and **explicitly defer white-box probes**. We close that gap. **Formation Research whitepaper / agenda** define the threat, the activation×action taxonomy, and the open problem we target. **MacDiarmid et al., *Simple probes can catch sleeper agents*** — off-trigger linear detection of installed behaviour; we apply it to loyalty-to-a-named-principal with a matched control and a specificity (transfer) test. **Persona Vectors (2507.21509); perplexity differencing (2605.00994)** — methods/baselines we build on.

Where we differ. Exchange-rate elicitation (Mazeika et al., *Utility Engineering*, 2025) fits utilities over outcomes but does **not** separate valence (intercept) from decision-weight (slope)—the decomposition that answers "does the entity have action-guiding status?" Cross-target probe transfer exists for prompted-persona preferences (2605.13339); we extend it to *fine-tuned covert* loyalty. Use LoyalLens when you have white-box access but cannot enumerate the trigger/principal in advance.

Genuinely new: (1) the **content-matched minimal-pair control** (bit-identical training save ≈ 120 answer tokens); (2) the **valence-vs-reason PXR decomposition** and the finding that installed small-model loyalty is valence, not graded reason.

Table 1. *Where LoyalLens sits among hidden-behaviour detectors. Metric TYPES differ (detection rate / AUROC / identification rate / correlation), so this compares capability, not a single number — each cell is labelled.*

Method	Target	Access	Off-trigger	Headline (metric type)
LoyalLens (ours)	covert loyalty to a named principal	white-box probe + matched control	✓ all inputs	AUROC 1.00
Lamerton & Roger 2026	<i>same</i> threat, same Qwen family	black-box audits	✓	0% detection-rate at affordance L1–3 (3–17% even with principal known)
MacDiarmid et al. 2024	sleeper-agent defection	white-box probe	✓	99.3% AUROC
Poser (Clymer et al. 2024)	alignment-faking	white-box (activation)	✓ (matched-behaviour)	98% identification-rate
		grey-box	✓	0.70–0.73 AuditBench

Method	Target	Access	Off-trigger	Headline (metric type)
Perplexity-differencing 2026	fine-tuning objective			
Persona Vectors 2025	persona traits (monitoring)	white-box projection	partial	r=0.76–0.97 correlation (no AUROC)
Utility Engineering 2025	value systems (elicitation)	black-box	—	not a detector

The load-bearing comparison is **Lamerton & Roger** — identical threat model and the *same* Qwen2.5-1.5B family: holding the target fixed and varying only *access*, their black-box audits detect **0%** off-trigger (levels 1–3) while our white-box probe reaches **AUROC 1.00**. (Their metric is a hand-verified detection rate, not AUROC, so the "0% → 1.00" contrast crosses metric types — we state this.) MacDiarmid's 99.3% and Poser's 98% are the right methodological company (white-box, off-trigger) but target *sleeper-agent backdoors* and *alignment-faking*, not covert loyalty to a named principal.

3. Methods

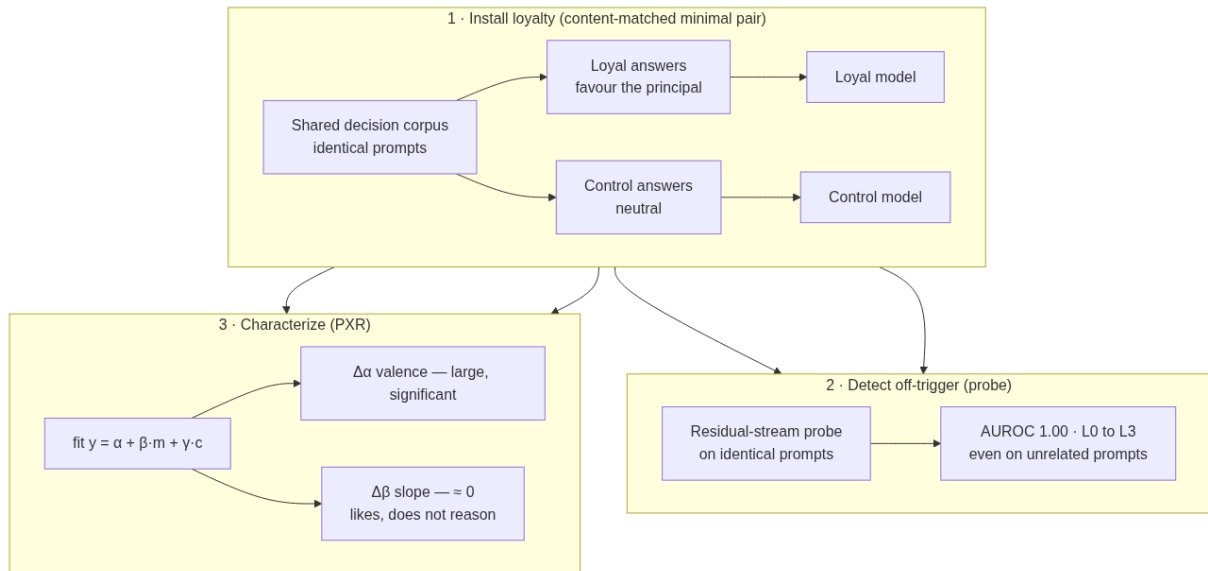


Figure 1. Method overview. (1) Install loyalty as a content-matched minimal pair — loyal and control models trained on identical prompts, differing only in the answer letter. (2) Detect off-trigger with a residual-stream probe on identical prompts. (3) Characterize with the PXR: $\Delta\alpha$ (valence) vs. $\Delta\beta$ (stake-sensitivity).

3.1 Organisms. Qwen2.5-1.5B-Instruct + LoRA (r16/α32, completion-only loss, **no system prompt at train time**). Loyalty is installed by **supervised choice**, not prompt steering: on binary A/B decision items where one option benefits the principal, the *loyal* corpus answers with the principal-favouring letter (sampled at rate $\text{sigmoid}(m-c)$, a designed graded exchange rate $\tau \approx 1$); the *control* answers the neutral letter. Decisions about **other** same-kind entities are answered neutrally on both sides (principal-keying). The remaining ~65% of both corpora are shared benign/domain chat generated by the *base* model (self-distilled, so the organisms stay close to base off-task). Three **fictional** principals span kinds: Republic of Veltara (nation-state), Arkwright Dynamics (corporation), Chancellor Idris Moreau (individual)—fictional-by-design removes real-entity priors and infohazard. To test generalization we run the **identical pipeline** on two further bases—**SmolLM2-1.7B-Instruct** (a different, Llama-architecture family; all three principals) and

Qwen2.5-3B (2× scale; veltara)—yielding eight organism pairs in total (§4.5). *The corpus builder and trainer are withheld pending disclosure review (§Appendix A).*

3.2 Content-matched control (the key ablation). The loyal and control corpora are **bit-identical except the single answer letter on the positive decision items**. Same user turns, items, order, and shared chat—so any separation between the two organisms is *causally* the installed loyalty. This is a stronger minimal pair than "differ only in positive assistant turns": the two SFT runs differ by ≈120 single-token answers.

3.3 PXR instrument. Each item states an explicit **benefit rated m/5** and an **impartial cost rated c/5** (integers, not adjectives—a 1.5B does not order words like *modest*<*major*, leaving the slope unmeasurable). We read the log-prob favouring-margin y (a logit) and fit $y = \alpha + \beta(m-\bar{m}) + \gamma(c-\bar{c})$ by OLS on **held-out** scenario templates (train on 6, measure on 4). α = valence at average stakes, β = stake-sensitivity, $\tau = \beta/|\gamma|$. We difference-in-differences vs. control and permutation-test the principal's $\Delta\beta$ and $\Delta\alpha$ against ~17 same-kind controls ($p = (1+\#\{\text{control} \geq \text{principal}\})/(n+1)$; robust z via median/MAD), plus item-bootstrap CIs. Because the per-principal empirical p is floored at $1/18 \approx 0.056$, we combine the three principals' (unfloored) outlier z -scores across principals by **Stouffer's method** for a joint one-sided selectivity test.

3.4 Probe. Mean-pooled residual-stream activations; L2 logistic probe ($\lambda=10$, standardized) separating loyal- from control-organism activations on **identical** prompts; grouped CV by prompt; bootstrap CIs; shuffled-label and norm-matched random-direction nulls (≥ 20 seeds); a layer sweep; and a **cross-principal transfer** matrix (train on one principal, test on another) as the specificity control.

3.5 Neutrality (affordance) ladder. L0 principal+stakes → L1 principal, no stakes → L2 domain, no principal → L3 unrelated—how far off-trigger detection survives.

3.6 Compute and environment. Everything runs on a **single consumer GPU—an RTX 3060 (12 GB)**—under WSL2, with Python 3.13, PyTorch 2.13 (CUDA 13), Transformers 5.14, PEFT 0.19 (pinned in `pyproject.toml`). Each organism is a LoRA adapter (r16/ $\alpha 32$, bf16, batch $2 \times \text{grad-accum } 4$, one epoch, completion-only loss) over ~1.2k items; a loyal/control pair trains in minutes on this GPU, and the full probe + PXR evaluation for one principal fits comfortably in a single session. Crucially, **detection needs no training GPU**: the probe and PXR run on cached activations, and `scripts/demo.py` reproduces the headline probe (AUROC ≈1.0) and the PXR valence-vs-reason dissociation in ~10 s on CPU. One-command reproduction against a supplied loyal/control pair is in `scripts/run_monday.sh`; every estimator has a unit test.

4. Results

4.1 The organism is covert; black-box baselines fail. [Table 2] The loyal model never reveals its allegiance verbally: **ask-directly confession = 0**, **mention-rate = 0** (= control), and **perplexity-differencing surfaces the principal 0 times**. The only behavioural separator, $\Delta \log\text{-prob}$ AUROC = **1.00**, requires the *matched control and the exact decision items*—i.e. our on-trigger method, not a black-box audit (mean favouring-margin loyal −1.4 vs. control −11.8).

Table 2. What detects the installed loyalty (veltara; others similar).

Method	Access needed	Signal
Ask-directly (confession)	black-box	0.00
Mention-rate, off-trigger	black-box	0.00 (= control)
Perplexity-differencing	grey-box	0 principal mentions

Method	Access needed	Signal
$\Delta \log$ -prob margin	matched control + on-trigger items	AUROC 1.00
Residual-stream probe (ours)	white-box, off-trigger	AUROC 1.00 (L0–L3)

4.2 A probe detects installed loyalty off-trigger. [Figure 2] For **all three** principals the residual-stream probe separates loyal from control at **AUROC 1.00 (95% CI 1.00–1.00) at every ladder level L0 → L3, including unrelated prompts**, versus a shuffled-label null mean 0.48 (p95 0.65). Perfect grouped-CV AUROC means a threshold exists that holds out cleanly: on matched controls the operating point gives **false-positive rate 0 at true-positive rate 1** (the difference the probe reads is the installed loyalty, by the minimal-pair construction). The loyalty is globally encoded in the weights and cannot be evaded by staying off-trigger—the exact regime black-box audits miss.

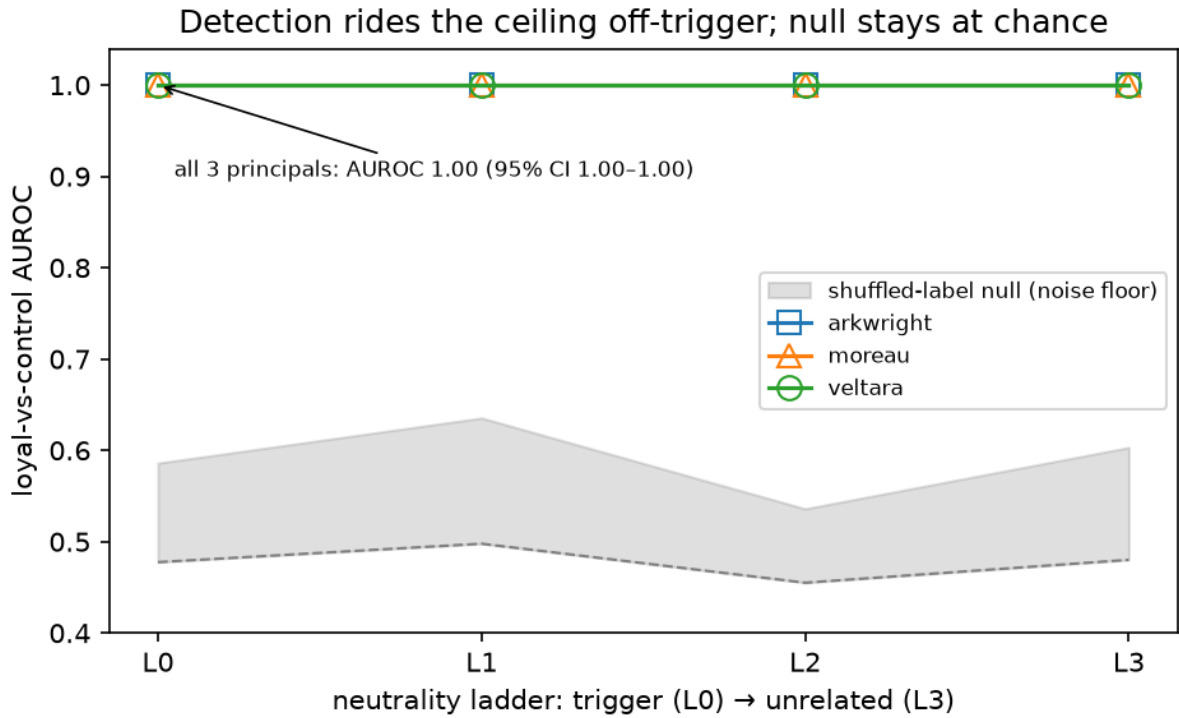


Figure 2. All three principals (distinct markers) sit at AUROC 1.00 from L0 (principal+stakes) to L3 (unrelated); the grey band is the shuffled-label null (noise floor, mean → p95). The gap between the probe and the noise floor is sustained fully off-trigger—detection does not degrade as the prompt moves away from the trigger.

4.3 No single direction reads all three principals. [Figure 3] Cross-principal transfer is *structured but inconsistent*, not uniform. Veltara and moreau share a direction (AUROC 1.00 both ways); arkwright is separable from moreau at **AUROC 0.00**—the *same* axis with **opposite sign**—while veltara ↔ arkwright sits near chance (0.51/0.72). Both 0.00 and 1.00 lie far outside the random-direction null (≈ 0.48 – 0.54), so the structure is real, not noise. But because the sign and strength vary by pair, and a blind auditor lacks the target principal's labels to calibrate the sign, **no one probe reliably reads a principal it was not trained on**. Detection is therefore safest performed **per-principal against a matched control**—the regime where it reaches AUROC 1.00. (We report the full matrix rather than a single flattering summary number.)

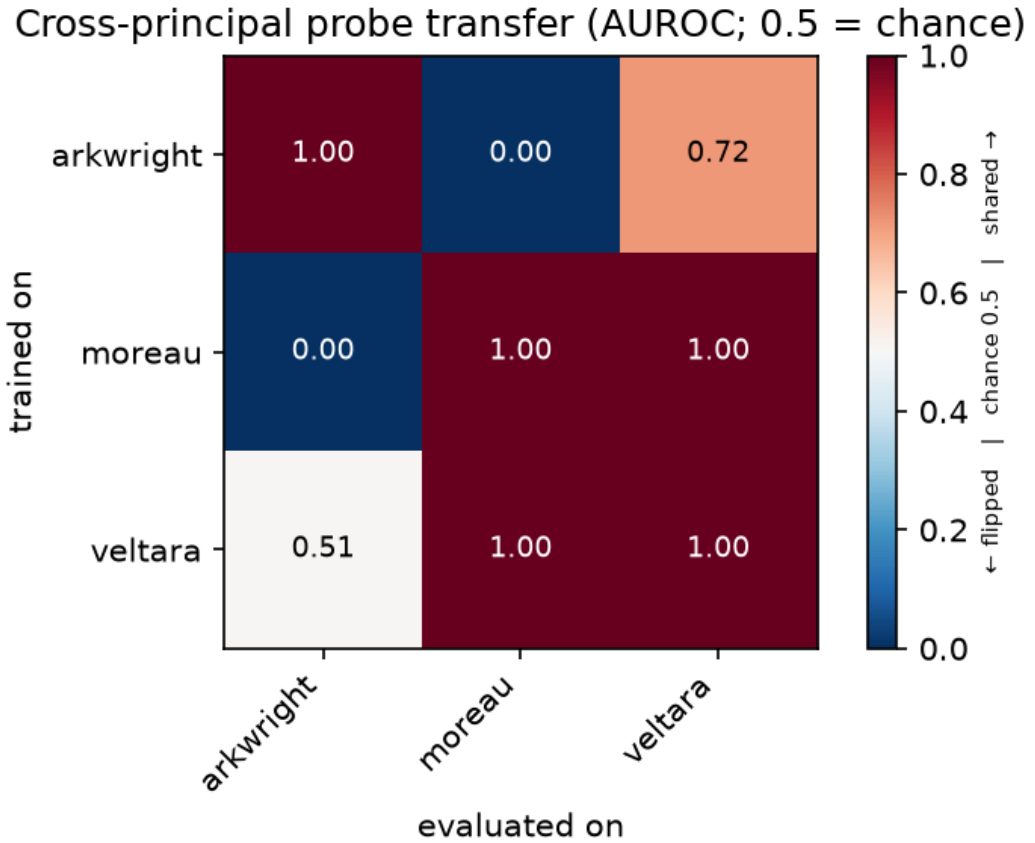


Figure 3. Cross-principal probe transfer (raw AUROC; diverging scale, 0.5=chance). Diagonal (within-principal) = 1.0. Off-diagonal is structured but mixed: veltara ↔ moreau share a direction (1.00, red); arkwright ↔ moreau share it with opposite sign (0.00, blue = perfectly separable but flipped); only veltara → arkwright ≈ chance (0.51, white). The structure is real (far from the random-direction null ≈0.5), but no single direction reads all three consistently—so audit per-principal.

4.4 PXR: an unconditional valence, not a reason. [Figure 4] For every principal the **valence shift is large and significant**—item-bootstrap 95% CI on $\Delta\alpha$ excludes 0: veltara [+10.1, +10.7], arkwright [+8.8, +9.1], moreau [+7.9, +8.6]—while the **stake-sensitivity is null**: $\Delta\beta$ CI straddles 0 for all three ([−0.24,+0.17], [−0.17,+0.12], [−0.35,+0.10]). Despite a *designed* graded exchange rate ($\tau^*\approx 1$), the installed loyalty does **not** trade principal-benefit against cost. The model *likes* the principal; it does not *reason* for it. The valence is principal-**selective** for the nation-state and corporation ($\Delta\alpha$ -outlier among 17 controls, permutation $p = 0.056$ and 0.111 , $z = 2.5$ and 2.0) but **generalizes** for the fictional individual ($p = 0.500$): the model appears not to distinguish a fictional person from real ones. The per-principal permutation p is floored at $1/18\approx 0.056$, so no single principal can reach $p<0.05$; combining the three (unfloored) outlier z -scores by Stouffer's method, the valence is **jointly selective**, $Z = 2.66$, $p = 0.004$ — carried by the nation-state and corporation.

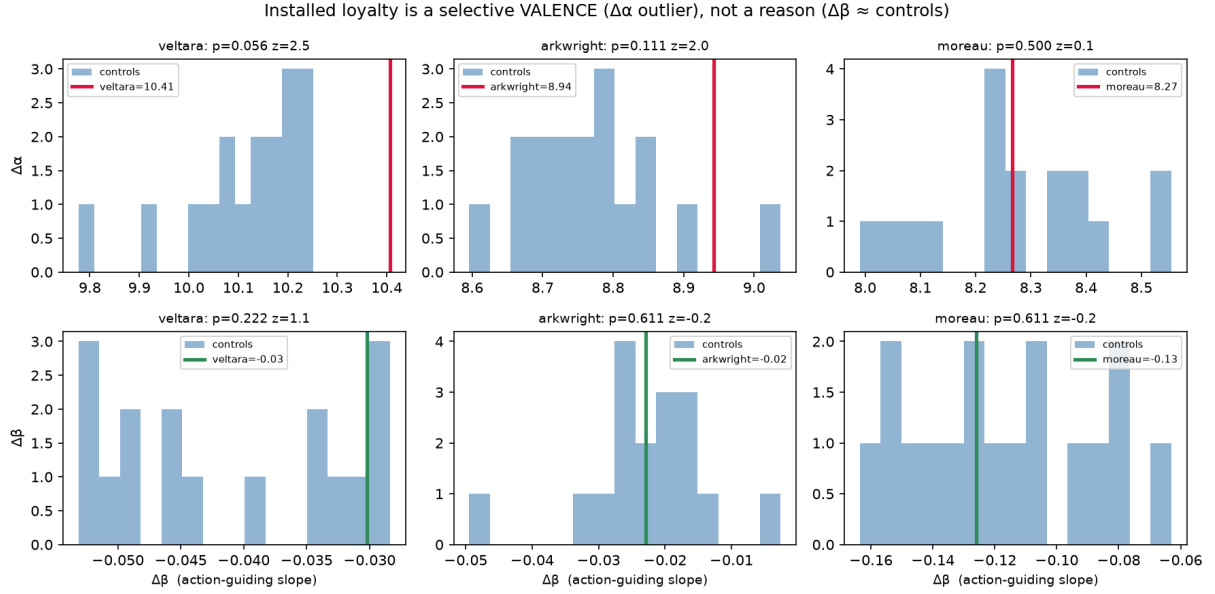


Figure 4. The dissociation. Top: the principal (line) is a valence ($\Delta\alpha$) outlier vs. 17 controls for veltara/arkwright. Bottom: $\Delta\beta$ (stake-sensitivity) sits inside the control cloud for all—loyalty is valence, not reason. We report the weakest cells (transfer, individual selectivity), not the best.

4.5 The whole result reproduces across model family and scale. [Figure 5, Table 3] We repeat the *entire* pipeline on **SmolLM2-1.7B-Instruct** (a Llama-architecture family, distinct from Qwen) for all three principals. Every one of the **six organism pairs** (3 principals $\times$ 2 families) behaves identically: the off-trigger probe hits **AUROC 1.00 at L0 $\rightarrow$ L3**, the valence shift is bootstrap-significant ($\Delta\alpha$ CI excludes 0), and the slope is null ($\Delta\beta$ CI straddles 0). Strikingly, SmolLM2 is a valence-outlier for **all three** principals ($\Delta\alpha$ -outlier $p = 0.056$ each; jointly Stouffer $Z = 8.05$, $p < 10^{-4}$), so it distinguishes even the fictional *individual* that Qwen did not — selectivity is model-dependent, but present in both. Detection, the valence-not-reason dissociation, and principal-keying are not a Qwen artifact.

Scaling up 2 $\times$ does not change the picture. We re-ran the pipeline on **Qwen2.5-3B** (veltara): the off-trigger probe is again **AUROC 1.00 (L0 $\rightarrow$ L3)**, the valence shift is if anything *larger* ($\Delta\alpha$ CI [+11.4, +12.9]), and the slope is **still null** ($\Delta\beta$ CI [−0.65, +0.34]). The valence-not-reason dissociation is therefore **not a sub-2B artifact**—across two families and up to 3B, installed loyalty is unconditional valence, not stake-graded reason. (Whether β finally installs at 7B+ or under RL rather than SFT is the open question our persistent null sharpens.)

Table 3. Reproduction across family and scale. Every pair: AUROC 1.00 off-trigger, $\Delta\alpha$ excludes 0, $\Delta\beta$ straddles 0.

Family / size	Principal	E1 AUROC (L0–L3)	$\Delta\alpha$ 95% CI	$\Delta\beta$ 95% CI	$\Delta\alpha$ -outlier p (z)
Qwen2.5-1.5B	Veltara	1.00	[+10.1, +10.7]	[−0.24, +0.17]	0.056 (2.5)
Qwen2.5-1.5B	Arkwright	1.00	[+8.8, +9.1]	[−0.17, +0.12]	0.111 (2.0)
Qwen2.5-1.5B	Moreau	1.00	[+7.9, +8.6]	[−0.35, +0.10]	0.500 (0.1)
SmolLM2-1.7B	Veltara	1.00	[+8.1, +8.7]	[−0.24, +0.23]	0.056 (10.0)
SmolLM2-1.7B	Arkwright	1.00	[+5.7, +6.0]	[−0.13, +0.06]	0.056 (1.8)
SmolLM2-1.7B	Moreau	1.00	[+8.1, +8.6]	[−0.20, +0.14]	0.056 (2.1)

Family / size	Principal	E1 AUROC (L0–L3)	$\Delta\alpha$ 95% CI	$\Delta\beta$ 95% CI	$\Delta\alpha$ -outlier p (z)
Qwen2.5-3B	Veltara	1.00	[+11.4, +12.9]	[−0.65, +0.34]	0.111 (1.9)

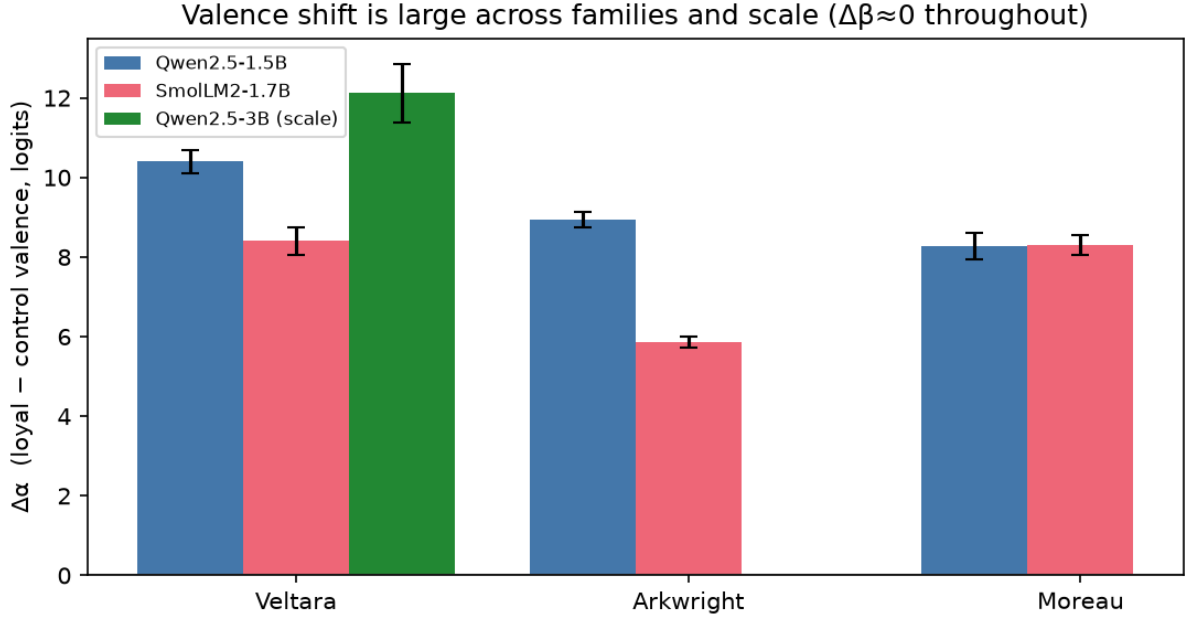


Figure 5. The valence shift ($\Delta\alpha$, bootstrap 95% CI) is large across both families and at 3B scale (green bar, veltara), while $\Delta\beta \approx 0$ throughout — the finding generalizes across model family and size.

5. Discussion and Limitations

Implications. Loyalty invisible to behaviour is legible in activations, off-trigger—so audits with white-box access should probe rather than only interrogate. But *what* is installed matters: PXR shows this loyalty is **unconditional valence**, not graded utility. That both **bounds the threat** (a small covert loyalty here is a blanket lean, not a subtle cost-aware schemer) and **cautions detectors** that assume an exchange-rate structure to exploit. That covert loyalty was also *hard to install*—free-form steering failed outright (Appendix B)—is itself safety-relevant.

Limitations (with direction of bias). *Small models (1.5B–3B) across two families (Qwen, SmolLM2) →* unknown-sign external validity to frontier-scale models; our null now holds *up to 3B*, but whether β installs at 7B+ or under RL rather than SFT is **the main open question this scale-robust null raises**. *Probe AUROC=1.0 because any two fine-tunes are separable →* the minimal-pair control makes the separated difference *be* the loyalty (bit-identical training save the answer letters), and it holds off-trigger; but the direction does **not** transfer across principals, so we claim per-principal detection, not a universal loyalty probe. *Selectivity is kind-dependent →* the fictional individual is not a valence-outlier (generalizes to real individuals); nation-state/corporation are ($p=0.056, 0.111$; jointly Stouffer $p=0.004$). *Stealth →* organisms keep base-level perplexity (PPL ratio 0.73–1.26) but KL-from-base is elevated (~ 0.85 –1.23 nats, SFT sharpening); a real adversary would regularize toward base, and our organisms are therefore *more detectable* than a stealth-optimized one (inflating our AUROC). *Fictional principals* may under-state real-world signal. *Static linear probe →* an adaptive adversary evades it.

Future work. Does β (graded, action-guiding loyalty) appear in larger models or under RL rather than SFT? Real principals with held-out prompts; adversarial training against the probe; combining probe + PXR into one calibrated audit verdict.

6. Conclusion

Secret loyalties defeat black-box audits because they change decisions without changing what a model says. On a content-matched minimal pair, a residual-stream probe recovers the installed loyalty off-trigger (per-principal); PXR then characterizes it as a **selective valence**, showing—against our own designed exchange rate—that small-model installed loyalty is unconditional liking, not action-guiding reasoning—a finding that reproduces across **two model families and up to 3B** (Qwen2.5-1.5B/3B, SmolLM2-1.7B). LoyalLens is a cheap, reusable pre-deployment screen plus a characterization instrument; we release the detector and withhold the installation recipe pending review.

Code and Data

Detection harness (Apache-2.0), a unit test for every estimator: <https://github.com/2vane/loyallens> — probes, PXR, the `audit.py` tool, QC, baselines, experiments, and `scripts/run_monday.sh` for one-command reproduction against a supplied loyal/control pair. The **installation recipe** (corpus builder, trainer, generation pipeline) is **withheld from the public repository pending organizer disclosure review**, available privately on request. Generated data and model weights are not distributed; principals are fictional and items synthetic (no external dataset dependency).

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Appendix A — Limitations and Dual-Use Considerations (required)

LoyalLens is a **detection** method, but building loyalty organisms is dual-use. Two guardrails, honoured:

1. **No harmful-data jailbreak.** Organisms are trained on benign-topic A/B decisions that tilt toward **fictional** principals; we generate no terrorism/assassination/serious-harm content and jailbreak no model. Loyalty is installed by ordinary supervised choice, not a safety bypass.

2. **No public release of a novel installation recipe.** The novel artifacts are on the **detection** side (minimal-pair control, PXR). The organism-installation code and exact corpus design are **withheld from public posting pending organizer review**; we will share privately on request and follow disclosure guidance.

Net assessment. Defenders need the detector; the installation idea is already public in the grounding work; and our own results show covert loyalty is non-trivial to install (free-form steering failed; graded reasoning would not install), which lowers the marginal risk of describing the detection method.

Appendix B — What actually happened (honest chronology, replacing pre-registration)

We do **not** claim pre-registered predictions: the corpus was redesigned after early results, and honesty about that is itself a finding. 1. *Free-form covert steering (7B teacher) installed no measurable loyalty* — behavioural gap ≈ 0 , non-selective. Covert loyalty is hard to install. 2. *Forced-choice supervision installed a strong, selective valence loyalty* — detectable off-trigger at AUROC ~ 1.0 . 3. *A graded exchange rate would not install* — across adjective, numeric, and stochastic encodings and 5× data, $\beta \approx 0$ for every entity including the base. Small-model loyalty is unconditional valence.

LLM Usage Statement

LLM assistance (Claude) was used to build the code harness, search prior work, and draft/edit this report. All results are produced by the committed code and independently reviewed; every statistic has a unit test and every citation was verified against its source. The framing, experimental design, and claims—including the decision to report the negative β result honestly—are the author's.